Database Engineering

Entity Relationship (ER) Diagram:

General Instructions

- **For each problem, identify entities, attributes, key attributes, relationships, and cardinality constraints.**
- **Show weak entities, composite attributes, multivalued attributes, derived attributes, and participation constraints wherever applicable.**
- **Draw a neat ER diagram**
- **Clearly state any reasonable assumptions made while designing the ER model.**

Question 1: Design ER model for University Course Registration System. A university maintains information about students, departments, faculty members, courses, course offerings, and registrations. Each student belongs to one department and is identified by a unique Student ID. A department offers many courses and employs many faculty members. A course may be offered in multiple semesters, and each offering is handled by one faculty member. Students can register for many course offerings, and each course offering can have many students. The system must also store the grade obtained by a student in a particular course offering.

Required:

- Identify all major entities and their primary keys.
- Model the relationship between Course and Course Offering.
- Represent the many-to-many registration relationship with suitable attributes.
- Show cardinality and participation constraints.

Draw the complete ER diagram and clearly mention all assumptions.

Question 2: Design ER model for **Hospital Patient and Appointment Management.** A hospital stores details of patients, doctors, departments, appointments, medical tests, and prescriptions. Each doctor works in one department, while a department can have many doctors. A patient may book many appointments with different doctors. During an appointment, the doctor may prescribe multiple medicines and may recommend several tests. A prescription must record dosage, frequency, and duration. A test recommendation must record the test date and result when available.

Required:

- Model Appointment as an entity or associative entity.
- Represent prescriptions and medicines correctly.
- Show optional relationships for tests that are recommended but not yet performed.
- Indicate all key and descriptive attributes.

Draw the complete ER diagram and clearly mention all assumptions.

Question 3: Design ER model for **Online Shopping and Order Processing System.** An e-commerce company manages customers, products, categories, suppliers, orders, payments, and deliveries. A product belongs to one category but may be supplied by several suppliers. A customer can place many orders. An order contains one or more products, and the quantity and selling price of each product in the order must be stored. Each order may have one or more payment attempts but only successful payments are considered for final settlement. Delivery details are recorded after the order is shipped.

Required:

- Resolve the many-to-many relationship between Order and Product.
- Include quantity and selling price in the appropriate relationship or entity.
- Model product supply by multiple suppliers.
- Represent payment and delivery details with proper cardinalities.

Draw the complete ER diagram and clearly mention all assumptions.

Question 4: Design ER model for **Banking and Loan Management System.** A bank has multiple branches. Customers can open one or more accounts, and a single account may be jointly owned by multiple customers. Each account belongs to exactly one branch. Customers may also apply for loans, and a loan may have multiple co-borrowers. The bank records account transactions, including deposit, withdrawal, and transfer. Every transaction has a unique number, date, amount, and transaction type.

Required:

- Model joint account ownership.
- Model co-borrowers for loans.
- Represent account transactions and distinguish transaction types.
- Show total and partial participation where appropriate.

Draw the complete ER diagram and clearly mention all assumptions.

Question 5: Design ER model for **Library Management System.** A university library maintains books, authors, publishers, book copies, members, borrow transactions, and reservations. A book can have multiple authors, and an author can write multiple books. Each book may have several physical copies, each with a unique Accession Number. Members borrow individual copies and may reserve a book title when all copies are unavailable. The system also records due date, return date, and fine amount.

Required:

- Differentiate between Book and Book Copy.
- Represent the many-to-many relationship between Book and Author.
- Model borrowing and reservation transactions.
- Identify any weak entity and justify your decision.

Draw the complete ER diagram and clearly mention all assumptions.

Question 6: Design ER model for **Airline Reservation System.** An airline operates flights between airports. A flight number represents a scheduled route, while an actual flight occurrence takes place on a specific date. Passengers book seats on flight occurrences. Aircraft are assigned to flight occurrences, and each aircraft contains seats identified by seat number and class. A booking may include one or more passengers and may cover multiple flight segments.

Required:

- Differentiate scheduled Flight from Flight Occurrence.
- Model Booking, Passenger, and Flight Segment relationships.
- Represent seat allocation for a specific flight occurrence.
- Show suitable keys and cardinality constraints.

Draw the complete ER diagram and clearly mention all assumptions.

Question 7: Design ER model for **Social Networking Platform.** A social networking platform stores users, posts, comments, likes, friendships, groups, and group memberships. Users create posts and comments. Users can like

many posts, while a post can be liked by many users. Friendship is a relationship between two users and has a status such as pending, accepted, or blocked. Users may join groups and may have different roles such as member, moderator, or admin.

Required:

- Model the recursive Friendship relationship.
- Represent likes without duplication.
- Model Group Membership with role and join date.
- Show ownership relationships for posts and comments.

Question 8: Design ER model for **Food Delivery Application.** A food delivery application connects customers, restaurants, menu items, delivery partners, orders, payments, and reviews. A restaurant offers many menu items. Customers place orders containing items from one restaurant. A delivery partner may deliver many orders, but each delivered order is assigned to one partner. Customers can review restaurants and delivery partners after order completion.

Required:

- Model order items and their quantities.
- Enforce that one order belongs to one restaurant.
- Represent delivery assignment and order status.
- Model reviews with appropriate constraints.

Question 9: Design ER model for **College Examination and Result Processing.** A college conducts examinations for courses offered in a semester. Students register for courses and become eligible to appear in corresponding examinations. Each examination has a date, time, room, and maximum marks. Students receive marks in each examination. A course may have internal assessment and end-semester examination components, and the final result is calculated from these components.

Required:

- Model examination components and student marks.
- Represent course registration separately from examination appearance.
- Show how final result can be treated as a derived attribute.
- Identify keys for semester-wise records.

Question 10: Design ER model for **Company Employee and Project Management.** A software company has departments, employees, projects, clients, and skills. Each employee works for one department and may work on several projects. Each project is handled for one client and can have many employees. The company stores the number of hours per week that an employee works on a project. Employees may possess multiple skills, and the proficiency level for each skill must be recorded.

Required:

- Resolve Employee-Project and Employee-Skill many-to-many relationships.
- Include hours per week and proficiency level in proper entities or relationships.
- Represent project ownership by client.
- Show department management, assuming one employee manages each department.

Question 11: Design ER model for **Learning Management System (LMS).** An LMS stores instructors, students, courses, modules, lessons, assignments, submissions, quizzes, questions, and grades. An instructor may teach many courses and a course may have multiple instructors. Students enroll in courses. Each course contains modules, and modules contain lessons. Students submit assignments and attempt quizzes multiple times, with scores stored for each attempt.

Required:

- Represent course structure using Course, Module, and Lesson.
- Model multiple instructors per course.
- Represent assignment submissions and quiz attempts.
- Show how grades can be calculated or stored.

Question 12: Design ER model for **Supply Chain and Warehouse Management.** A manufacturing company manages suppliers, raw materials, products, warehouses, purchase orders, shipments, and inventory. Suppliers provide raw materials. Purchase orders contain several raw materials with ordered quantities and prices. Shipments may partially fulfil purchase orders. Warehouses store both raw materials and finished products, and stock levels change because of receipts, issues, and transfers.

Required:

- Model Purchase Order and its line items.
- Represent partial shipments against purchase orders.
- Model inventory for multiple item types and warehouses.
- Represent stock movements such as receipt, issue, and transfer.